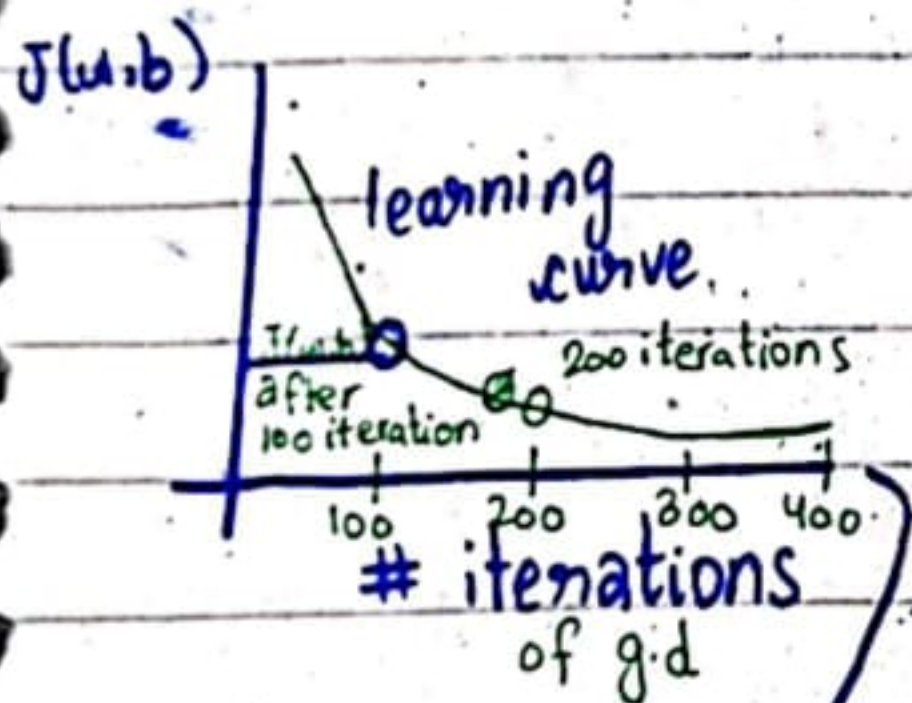


Checking Gradient Descent for Convergence

Remind: (Gradient Descent):

$$\begin{cases} w_j = w_j - \alpha \frac{\partial J(\vec{w}, b)}{\partial w_j} \\ b = b - \alpha \frac{\partial J(\vec{w}, b)}{\partial b} \end{cases}$$

objective: $\min_{w, b} J(\vec{w}, b)$



if g.d is working properly $J(w, b)$ should decrease after every iteration.

↳ if increase either alpha chosen poorly or bug in the code.

more likely the gradient descent has converged bcz curve no longer decreasing.

iterations varies could be 30, 1000, 10,000

difficult to tell in advance that how many iterations gradient descent needs to converge.

→ You can make learning curve in graph to see when to stop training the model.

Other method is:

Automatic convergence Test

Let ϵ "epsilon" be 10^{-3} .

If $J(\vec{w}, b)$ decreases by $\leq \epsilon$ in one iteration, declare "convergence".

(found parameters $\vec{w}, b$ to get close to global minimum).

